## ABSTRACT OF THE DISCLOSURE

A method and system for software-defined network routing and multiplexing over a massively parallel optical interconnect. The method treats each channel in a two-dimensional emitter array (e.g., each pixel in a micro-display) as an addressable data conduit. Routing information is encoded by mapping data packets to specific emission characteristics of the channels. For example, data intended for different network destinations can be assigned to different wavelengths (colors). A processor at the transmitting end acts as a network switch, analyzing incoming data and generating a frame where the color of each pixel determines the destination of the data it carries. A receiver decodes the frame and routes the data from each pixel based on its color. This technique allows a single physical interconnect to simultaneously carry multiple, logically separated data streams to different destinations, creating a high-density, software-defined network fabric.